

TEAM INN08

Real-Time Sign Language to Speech AR Glasses

AR glasses that give voice to 18 million deaf Indians — **offline,
hands-free, private.**

Mentor: **Prof. Dr. Pariza Kamboj**

DR Dwarkesh R.

JG Jenil G.

JP Jenny P.

AS Akifa S.

DM Drashtant M.

Business Overview

OUR MISSION

To give **voice to 18 million deaf Indians** through affordable, AI-powered AR wearables — eliminating the communication barrier that isolates them daily.

The Problem

-  India has only 339 certified ISL interpreters for 63M people with hearing loss
-  Deaf patients face misdiagnosis risk — no interpreters in 99% of hospitals
-  2–3× higher unemployment rate due to communication barriers
-  Existing apps require phones — hands are busy while signing

Our Solution

-  AI-powered AR glasses with real-time ISL-to-speech translation
-  Fully on-device processing — no internet needed, completely private
-  ₹15,000 retail (100% free under ADIP govt subsidy for BPL users)
-  Built exclusively for Indian Sign Language (ISL) — unlike any global competitor

BUSINESS TYPE

Hardware +
SaaS

REVENUE MODEL

B2C / B2G /
B2B

KEY CHANNEL

ADIP Govt
Scheme

YEAR 1 TARGET

50 Units

REGISTERED

Gujarat, India

Month 1-6

- Pilot at Indira Bharati & Muk Badhir Bal Vikas Kendra (Surat) — 20 students
- Collect user testimonials, accuracy data, and clinical evidence
- Apply for ADIP empanelment via DEPwD (scheme approved till Mar 2026)
- BIS certification for assistive device classification under RPwD Act

Month 7–18

- Launch direct website sales (₹15,000 retail)
- Partner with deaf associations for referral program
- Target urban professionals & students (highest purchase intent)
- Instagram/YouTube content with deaf community influencers

Year 2+

- Sell to inclusive employers (Amazon, Lemon Tree Hotels)
- Hospital compliance packages (RPwD disability mandates)
- State government bulk orders via ADIP at scale
- Educational institution licensing program

Our ₹15,000 device fits within ADIP's ₹15K full subsidy ceiling. For income $\leq ₹22,500/\text{mo}$ $\rightarrow$ 100% free. ₹22.5K–₹30K $\rightarrow$ 50% subsidy. Gujarat utilizes 75–80% of scheme funds.

Competitive Analysis & Advantages

Feature	Assistive Vision	SignAble App	Smart Gloves	XRAI Glass	Human Interpreters
ISL Recognition	✓	✓	✓	✗	✓
Hands-Free	✓	✗	✗	✓	✓
Offline / On-Device	✓	✗	✓	✗	✓
Real-Time (<500ms)	✓	✗	✓	✓	✗
Under ₹15,000	✓	✓	✗	✗	✗
ADIP Subsidized	✓	✗	✗	✗	✗
India-Specific ISL	✓	✓	✗	✗	✓



Hardware Patent

Filed patent on unique wearable form factor — structural protection against copycats



ISL Training Data

Proprietary 50K+ ISL gesture dataset — data is scarce (only 339 interpreters in India)



ADIP Empanelment

Govt scheme approval creates B2G distribution lock-in at zero customer acquisition cost



Community Network

Deep partnerships with Gujarat State Deaf Association and Surat schools

Only solution combining ISL recognition + hands-free AR + offline AI + ADIP subsidy eligibility — all at ₹15,000.

63M

1.9L

50

\$2.5B India Assistive Tech market (2025)

16% India wearable tech CAGR

\$772M India audiology devices (2024)

Our moat: ISL-specific, on-device AI, built for the ₹15K ADIP subsidy envelope.

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Assistive Vision — The Solution



Hands-Free OS

Wearable lightweight AR glasses — both hands remain entirely free for natural Indian Sign Language signing.

Seamless Real-Time Interaction



1. The User Signs

Deaf individual simply performs ISL gestures naturally in front of the hearing person.



2. Glasses Process

The built-in camera captures movements. On-board AI translates gestures into spoken words.



3. Instant Output

Sentences are spoken aloud via bone-conduction transducers instantly bridge the gap.

✓ Validation: In 15 interviews with deaf students & educators in Surat, 80% expressed daily usage intent.

Product Deep Dive

Hardware Stack

- **Raspberry Pi Zero 2 W** — Quad-core ARM, on-device AI brain
- **Pi Camera v2 (IMX219)** — 30 FPS gesture capture, 160° FOV
- **OLED Microdisplay** — HUD text confirmation overlay
- **Bone Conduction Audio** — Speech output — ears remain free
- **2500mAh LiPo Battery** — 6-8 hours continuous use
- **TR-90 Lightweight Frame** — Looks like regular glasses (~80g)

Current Prototype Results

Recognition Accuracy **85%**
Target: 90%+ in 6 months

Processing Latency **<500ms**
Target: <300ms

Battery Life
Target: 8 hrs

6.5 hrs

Device Weight
Target: 65g **~80g**

Signs Recognized
Target: 200 signs

50 ISL



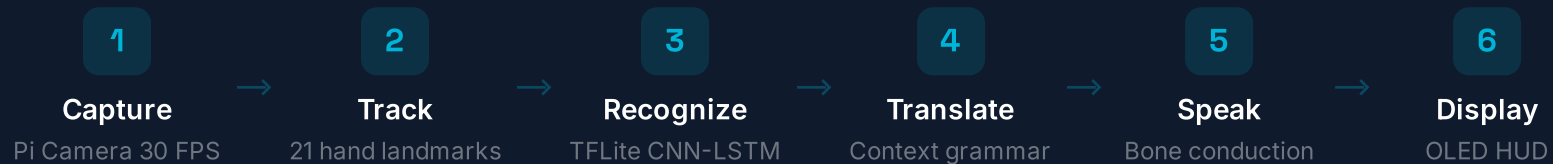
Live Demo

Watch as we sign "Hello, my name is Dwarkesh" and the glasses speak it aloud in real-time.

2-5 Min Live Demonstration

Validated by 15 deaf users in
Surat — 80% expressed daily
usage intent

End-to-End Pipeline — Under 500ms



Deaf Students

Less than 5% receive education in ISL. Communication gap with hearing teachers limits academic growth.

Working Professionals

Face a 2–3× higher unemployment rate. Cannot participate seamlessly in corporate meetings without interpreters.

Healthcare Centers

Critical misdiagnosis risks. 99% of hospitals do not have on-call ISL interpreters for deaf patients.

Rural Population

Complete communication isolation.
Cannot afford expensive
smartphones or subscription
interpreter apps.



The Rights of Persons with Disabilities (RPwD) Act explicitly mandates barrier-free accessibility in government institutions, hospitals, and educational facilities. **Currently, no scalable technology exists to help them comply.** Our B2B distribution taps directly into this mandated compliance budget.



The Ministry of Social Justice provides up to ₹15,000 for assistive devices under the ADIP scheme (approved until March 2026).

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The Team Behind the Vision

PK

Prof. Dr. Pariza Kamboj — Faculty Mentor

DR

Dwarkesh Ramani

Team Lead & AI/ML Engineer

"Built end-to-end gesture recognition pipeline with 85% accuracy"

Python TensorFlow

MediaPipe

System Architecture

JG

Jenil Gajera

Hardware & IoT Lead

*"Designed compact wearable
with 8-hour battery life at
80g"*

Raspberry Pi

Embedded Systems

PCB Design 3D Printing

JP

Jenny Patel

Frontend & UX Designer

*"Created intuitive interfaces
for both deaf and hearing
users"*

UI/UX Design

Accessibility User Research

AS

Akifa Shaikh

Business Development

*"Mapped ADIP subsidy
pathway for 100% user cost
coverage"*

Market Research

Financial Modeling

ADIP Scheme	Partnerships
1. Local Authorities 2. Healthcare Providers 3. Academic Institutions 4. Private Sector 5. Community Groups 6. Government Departments 7. Non-Profit Organizations 8. Research Centers 9. Industry Associations 10. International Organizations	1. Local Authorities 2. Healthcare Providers 3. Academic Institutions 4. Private Sector 5. Community Groups 6. Government Departments 7. Non-Profit Organizations 8. Research Centers 9. Industry Associations 10. International Organizations

DM

Drashtant Mevada

Business & Strategy

"Benchmarked 19 global competitors — identified clear market gap"

Competitive Analysis

Go-to-Market Documentation

Compliance

Financial Projections

ARR (YEAR 3)

₹55L

Recurring revenue from 5,500 users

3-Year Revenue Projection

Metric	Year 1 (Pilot)	Year 2 (ADIP + B2C)	Year 3 (Scale)
Units Sold	50	500	5,000
Hardware Revenue	₹7.5L	₹75L	₹7.5Cr
Recurring Revenue (ARR)	₹0.5L	₹5L	₹55L
Total Revenue	₹8L	₹80L	₹8.05Cr
COGS	₹5.5L	₹45L	₹4.1Cr
Gross Margin	27%	40%	45%
Team Size	5 founders	5 + 3 hires	5 + 8 hires
Monthly Burn	₹15K	₹2.5L	₹8L
Cash Flow +ve	—	Month 9	Month 4

COST STRUCTURE (YEAR 2)

COGS (hardware)	₹45L	56%
Salaries (8 people)	₹18L	22%
Marketing & Sales	₹8L	10%
R&D / Testing	₹5.6L	7%
Operations & Legal	₹4L	5%

₹2,00,000

Projected Spend Allocation

Deliverable: 2 working wearable devices	37.5%
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Deliverable: Patent application filed	20%
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Deliverable: 200-sign model at 90% accuracy	15%
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Deliverable: 20+ user feedback reports 12.5%

Deliverable: Production-ready frame design	10%
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Deliverable: Risk buffer	5%
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Thank You

Communication is a fundamental right.
With ₹2L, we give voice to 18 million Indians — for the first time.

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 TEAM
Team Inno8

Mentor: **Prof. Dr. Pariza Kamboj** | SSIP Student Innovation Grant

- Dwarkesh Ramani
- Jenil Gajera
- Jenny Patel
- Akifa Shaikh
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